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Isolation, characterization and inter-relationship of phosphate solubilizing bacteria from the rhizosphere of sugarcane and rice

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ABSTRACT

Phosphate solubilizing bacteria (PSB) have the ability to solubilize insoluble phosphorus to make it available for plant roots to be absorbed. In this study, isolation of rhizobacteria and screening of their ability for phosphate solubilization, production of indole acetic acid (IAA), antagonistic activity against fungal pathogen and intrinsic antibiotic resistance was studied. In total, 15 isolates of PSB were found gram negative with rod shaped cells. Different levels of antibiotic resistance was seen against four antibiotics (Ampicillin, Kanamycin, Tetracycline and Streptomycin 25, 30, 30 and 10 μg/ml respectively) by Rhizobacterial isolates (Seven isolates were found to show antifungal activity against *Fusarium oxysporum*). All PSB isolates were found to solubilize insoluble phosphorous $Ca_3(PO_4)$ and produce IAA. Two PSB isolates sequences were found novel and were submitted in NCBI database. Conclusively, combine application of rhizobacterial isolates, Sugarcane (SC-22) in addition with antifungal agent can prove to be an excellent combination to solubilize insoluble phosphorus and to act as antifungal remedy. Further this combination can be eco-friendly and prove to be cost effective strategy to improve crop production.

1. Introduction

Phosphorus is considered to be one of the essential macro-element required for growth and development of plant (Kumar et al., 2010). Plants obtain phosphorus from the soil solution. However, due to low solubility and fixation in soils, only a small fraction of phosphorus exists in soil solution (1 ppm or 0.1%), which is readily available to plants. The roots take up several forms of phosphorus, out of which the greatest part is absorbed in the forms of $H_2PO_4^-$ and HPO_4^{2-} depending upon soil pH (Mahidi et al., 2011). Certain bacteria called PSB have the ability to solubilize insoluble phosphorus and make it available to the plant roots for absorption. PSB are ubiquitous whose numbers vary from soil to soil. PSB constitute 1–50% of the total respective population in soil (Chen et al., 2006). They are predominately concentrated in the rhizosphere, where they are known to be metabolically more active than those isolated from other sources (Vessey, 2003). Evidence of naturally occurring rhizospheric PSB dates back to 1903 (Kiani et al., 2013). Strains from bacterial genera *Pseudomonas*, *Bacillus*, *Rhizobium*and *Enterobacter* are the most powerful phosphate solubilizer (Fu et al., 2016; Wakelin et al., 2004; Zaidi et al., 2009). The major phosphate solubilizing mechanism involves the production of organic acids accompanied by acidification of the medium. The organic acids convert tricalcium phosphate to di- and mono- basic phosphates with the net result of an enhanced availability of the element to the plant. The type of organic acids produced and their amounts differ with different organisms. Among them, Gluconic acid and 2-ketogluconic acid seems to be the most frequent agents of mineral phosphate solubilization. Other organic acids, such as acetic, citric, lactic, propionic, glycolic, oxalic, malonic, succinic acid, fumaric, tartaric acids have also been implicated in phosphate solubilization (Ahmad and Sahab, 2011). Indoleacetic acid (IAA) is a molecule that is synthesized by plants and a few microbes. In plants, IAA plays a key role in both root and shoot development. The hormone moves from one part of the plant to another by a designated importer (AUX1) and efflux pumps (PIN1–7). Plant Growth promoting hormone (PGPR) help plants to synthesize this molecule for better growth and root developmentsSo this also a parameter to check wether PGPR are working or not (Sheikhian and Bina, 2016).

Global crop production is facing a serious threat of food security and stability because of the current yield losses caused by pathogenic microbs. Today's intensified agricultural practices utilize huge amounts of

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Table 1
Soil analysis of rhizospheric samples taken from the rhizosphere of sugarcane and rice.

S. no.	Location	Depth (in.)	EC (mS cm^{-1})	pH	Organic matter (%)	Available phosphorus (mg kg-1)	Available potassium (mg kg-1)	Saturation (%)	Texture
1	Gujranwala	0–12	3.50	8.3	0.63	2.80	93	36	Loam
2	Gujranwala	0–12	2.73	8.4	0.70	3.33	100	49.33	Loam
3	Shaikhupura	0–12	1.83	7.9	0.86	4.53	87.33	52	Clay
4	Shaikhupura	0–12	1.47	7.7	0.96	5.57	119.33	52	Clay
5	Gujranwala	0–12	3.23	7.6	0.84	9.30	105	52	Clay
6	Lahore	0–12	1.80	7.7	1.05	5.30	129	52	Clay
7	Lahore	0–12	3.20	7.6	0.90	12.04	111	52	Clay
8	Lahore	0–12	1.60	7.8	0.80	5.70	90	52	Clay

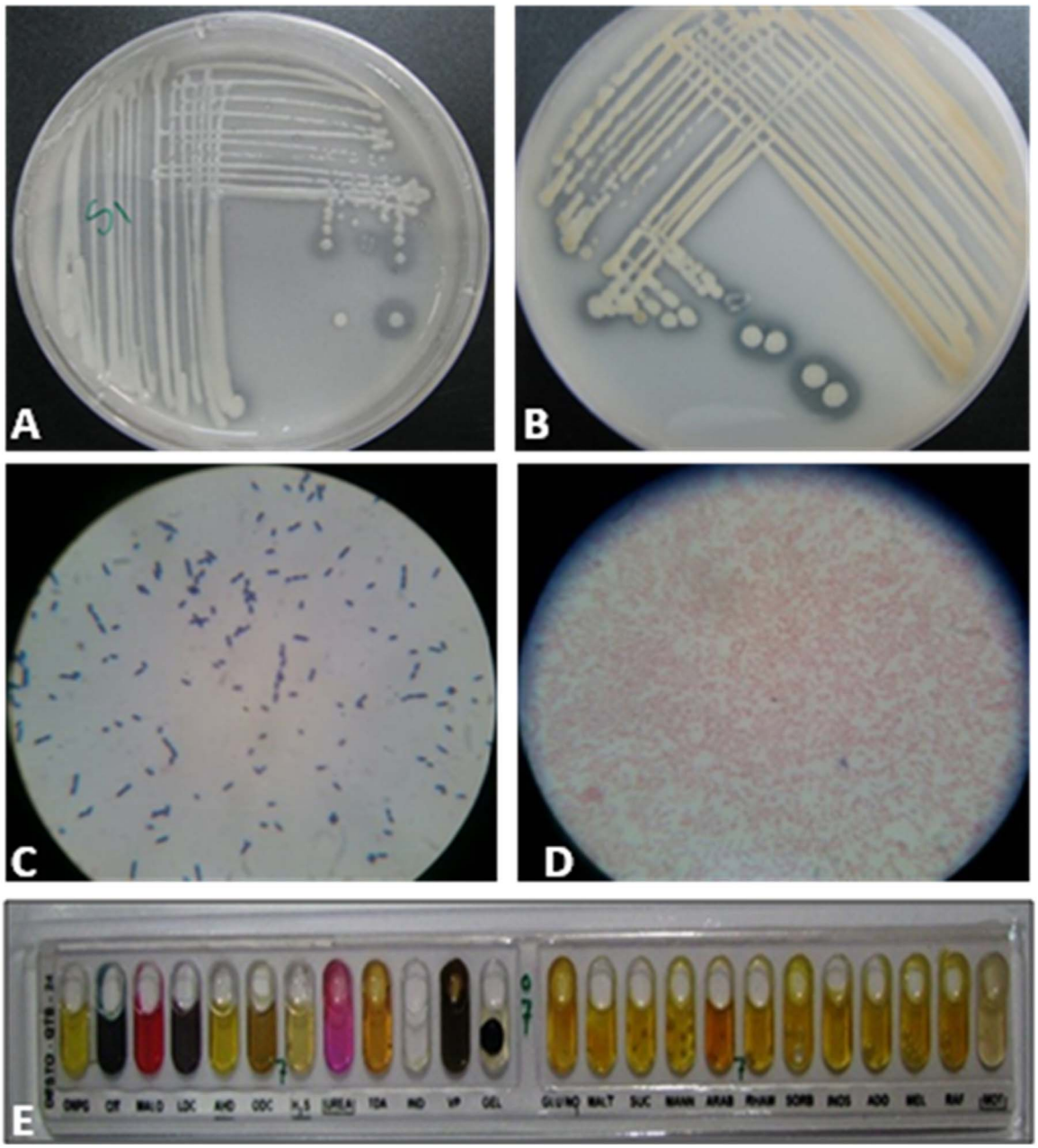

Fig. 1. Biochemical Identification of Rhizobacterial isolates. Purified PSB isolates identified on the basis of their (A, B): capability of forming clear zones of phosphate solubilization around colonies on PVK/NBRIP agar media, (C,D): Gram staining differentiating Rod shaped cells and round shaped cells, (E): QTS-24 for identification of phosphate solubilizing rhizobacterial isolates.

Table 2
Morphological characteristics of phosphate solubilizing rhizobacterial isolates.

Isolate name	Odor	Surface	Elevation	Growth in liquid medium	Margin	Color	Optical feature	Shape	Arrangement
R-1	Odorless	Shiny	Raised	Turbid	Smooth	Off white	Opaque	Rod	Scattered
R-2	Odorless	Shiny	Raised	Turbid	Wavy	Skin	Opaque	Rod	Scattered
R-3	Odorless	Mucoid	Raised	Turbid	Smooth	Off white	Opaque	Rod	Scattered
SC-01	Odorless	Mucoid	Raised	Turbid	Wavy	Off white	Opaque	Rod	Paired
SC-03	Odorless	Shiny	Raised	Turbid	Smooth	Skin	Opaque	Rod	Scattered
SC-05	Odorless	Shiny	Raised	Turbid	Smooth	Off white	Opaque	Rod	Scattered
SC-07	Odorless	Shiny	Raised	Turbid	Smooth	Off white	Opaque	Rod	Scattered
SC-08	Odorless	Shiny	Raised	Turbid	Smooth	Off white	Opaque	Rod	Scattered
SC-09	Odorless	Shiny	Flat	Turbid	Smooth	Off white	Opaque	Rod	Scattered
SC-13	Odorless	Shiny	Raised	Turbid	Wavy	Off white	Opaque	Rod	Scattered
SC-15	Odorless	Shiny	Raised	Turbid	Smooth	Off white	Opaque	Rod	Paired
SC-17	Odorless	Shiny	Raised	Turbid	Wavy	Off white	Opaque	Rod	Long Chain
SC-19	Odorless	Mucoid	Flat	Turbid	Wavy	Off white	Opaque	Rod	Scattered
SC-20	Odorless	Shiny	Raised	Turbid	Smooth	Skin	Opaque	Rod	Paired
SC-22	Odorless	Mucoid	Raised	Turbid	Wavy	Yellowish	Opaque	Rod	Long Chain

Table 3
Biochemical tests for the identification of Phosphate solubilizing rhizobacterial isolates.

Sr. no	Biochemical tests	Isolate name														
		R-1	**R-2**	**R-3**	**SC-01**	**SC-03**	**SC-05**	**SC-07**	**SC-08**	**SC-09**	**SC-13**	**SC-15**	**SC-17**	**SC-19**	**SC-20**	**SC-22**
1	Gram Test	Gram -	Gram -	Gram -	Gram -	Gram -	Gram -	Gram -	Gram -	Gram -	Gram -	Gram -	Gram -	Gram -	Gram -	Gram -
2	Catalase Test	+	+	+	+	–	+	+	+	+	+	+	+	+	+	+
3	Oxidase Test	+	–	–	+	–	+	–	+	+	–	–	+	–	–	–
4	Lactose Test	+	+	+	–	–	+	–	–	+	+	+	+	+	+	+
QTS-24 Tests																
1	ONPG	+	+	+	–	–	+	+	+	+	+	–	–	+	+	+
2	CIT	+	+	–	+	–	–	+	–	+	+	–	–	+	+	+
3	MALO	+	+	+	–	–	+	+	+	+	+	–	–	–	+	–
4	LDC	+	+	+	+	–	+	–	–	–	–	–	–	–	+	+
5	ADH	+	–	–	–	–	–	–	+	+	–	–	+	+	–	+
6	ODC	–	–	+	–	–	+	–	–	–	+	–	–	–	–	–
7	H_2S	+	–	–	+	+	–	–	–	–	+	–	–	–	–	–
8	UREA	–	+	–	–	+	–	–	–	–	–	–	–	–	+	+
9	TDA	+	+	–	+	–	+	+	+	–	+	+	–	+	+	+
10	IND	+	–	–	–	+	+	–	–	–	–	–	–	–	–	+
11	VP	+	+	+	–	+	–	–	+	+	–	–	–	+	–	+
12	GEL	+	–	+	+	–	–	+	+	–	+	–	–	+	–	+
13	GLU	+	+	+	+	+	–	–	+	–	+	–	+	+	+	+
	NO_3/N_2	+	+	+	–	+	+	–	–	+	+	–	–	–	+	–
14	MALT	+	+	+	+	+	+	+	+	+	+	–	–	+	+	+
15	SUC	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
16	MANN	+	+	+	–	–	+	–	+	+	+	–	–	+	+	+
17	ARAB	+	+	+	+	+	–	–	+	+	+	+	+	+	+	+
18	RHAM	+	+	+	+	–	+	+	+	+	+	+	+	+	+	+
19	SORB	–	+	+	–	–	+	+	+	+	–	–	–	–	+	–
20	INOS	–	+	+	–	–	+	+	+	+	–	–	–	+	+	+
21	ADO	–	–	+	–	–	+	–	+	+	–	+	–	–	–	–
22	MEL	–	+	+	–	+	+	+	+	+	–	–	–	+	+	+
23	RAF	–	–	+	–	–	–	+	–	+	–	–	–	–	–	–
24	MOT	+	–	+	+	+	–	+	–	+	+	–	+	+	–	–
25	CO	+	–	–	+	–	+	–	+	+	–	–	+	–	–	–

Microbial identification kits (QTS-24) were used for these Biochemical tests, ONPG = Ortho nitrophenyl β-D-galactopyranoside; CIT = Sodium citrate; MALO = Sodium malonate; LDC = Lysine decarboxylase; ADH = Arginine dihydrolase; ODC = Ornithine decarboxylase; H2S = H2S production; UREA = Urea hydrolysis; TDA = Tryptophan deaminase; IND = Indole; VP = (Vogesproskauer); GEL = Gelatin hydrolysis; GLU = Acid from glucose; NO3 = Nitrate reduction; N2 = N2 gas; MALT = Acid from maltose; SUC = Acid from sucrose; MANN = Acid from mannitol; ARAB = Acid from arabinose; RHAM = Acid from rhamnose; SORB = Acid from sorbitol; INOS = Acid from inositol; ADO = Acid from adonitol; MEL = Acid from melibiose; RAF = Acid from raffinose; MOT = Motility; CO = Cytochrome oxidase. Where, − = Stands for negative in test and + = Stands for positive in test.

agrochemicals for crop protection against these pathogens. However, increased use of chemicals has severe negative impacts on non-target pest (Dhandapani et al., 2011). Therefore, the use of biocontrol agents including bacteria seems to be a safe and attractive alternative that can replace the input of chemicals in agriculture sector. Rhizospheric microorganisms are reported to produce growth promoting substances in large quantities that have direct influence on plant morphology (Tariq et al., 2014; Richardson et al., 2001; Dar et al., 2014). Correlation analysis provides an opportunity to select better individuals on the basis of strength of association (Joe et al., 2016a, 2016b; Ali et al., 2013, 2014a, 2014b)

Keeping in view, the international concerns for food and environment quality, utilization of PGPR instead of chemicals in agriculture is an issue of significant importance. Many formulations of PGPRs have been reported that enhance growth, seed emergence and yield of crop plants and some of them have been commercialized as well. The current study is an attempt to isolate and characterize potential PSB from rhizospheric soil of sugarcane and rice. The correlation among the isolates was determined to find the association of using isolates in combination.

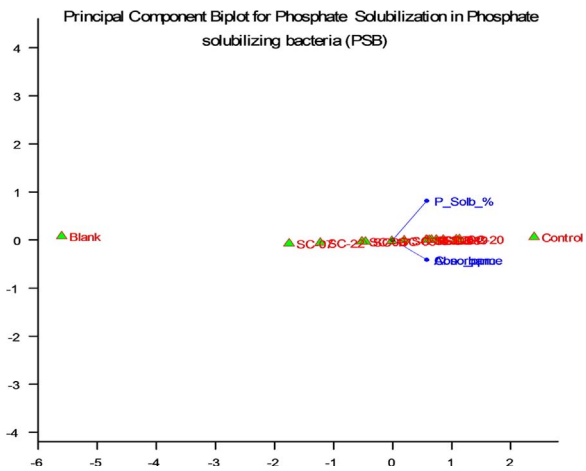

Fig. 2. Inorganic phosphate solubilization by PSB isolates.

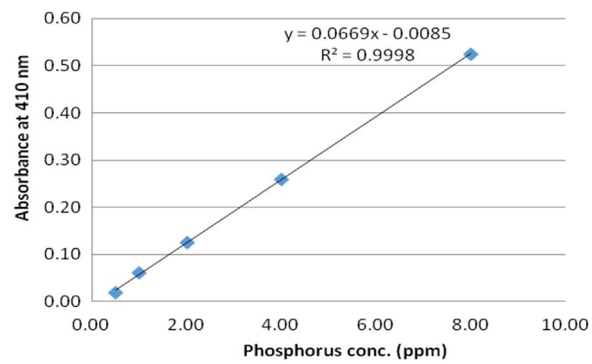

Fig. 4. Standard curve for phosphorus.

2. Materials and methods

2.1. Morphological and biochemical identification

For isolation of PSBs, 10 rhizospheric soil samples from Sugarcane (*Saccharum* Sp.) and rice (*Oryza sativa*) fields were collected from eight different locations of Gujranwala, Shaikhupura and Lahore cities (Punjab, Pakistan) during summer 2012. Soil samples were analyzed for electrical conductivity (EC), pH, organic matter content, available potassium, saturation and available phosphorus from Soil and Water testing Laboratory (Agriculture Department, Government of Punjab) (Analysis Report No. 2977/167).

Isolation of Phosphate Solubilizing rhizobacteria was carried out by spread plate technique. Dilutions of soil samples were prepared of the root homogenate and soil (10% soil in 0.85% saline). The well-developed, predominant, morphologically distinct single colonies with clear halo zones were selected and further purified by streaking on Pikovskaya's (1948) and NBRIP agar medium (Nautiyal, 1999a, 1999b). Morphological identification of the rhizobacterial isolates was done on the basis of colony and cell morphology (De-Freitas et al., 1997; Perez

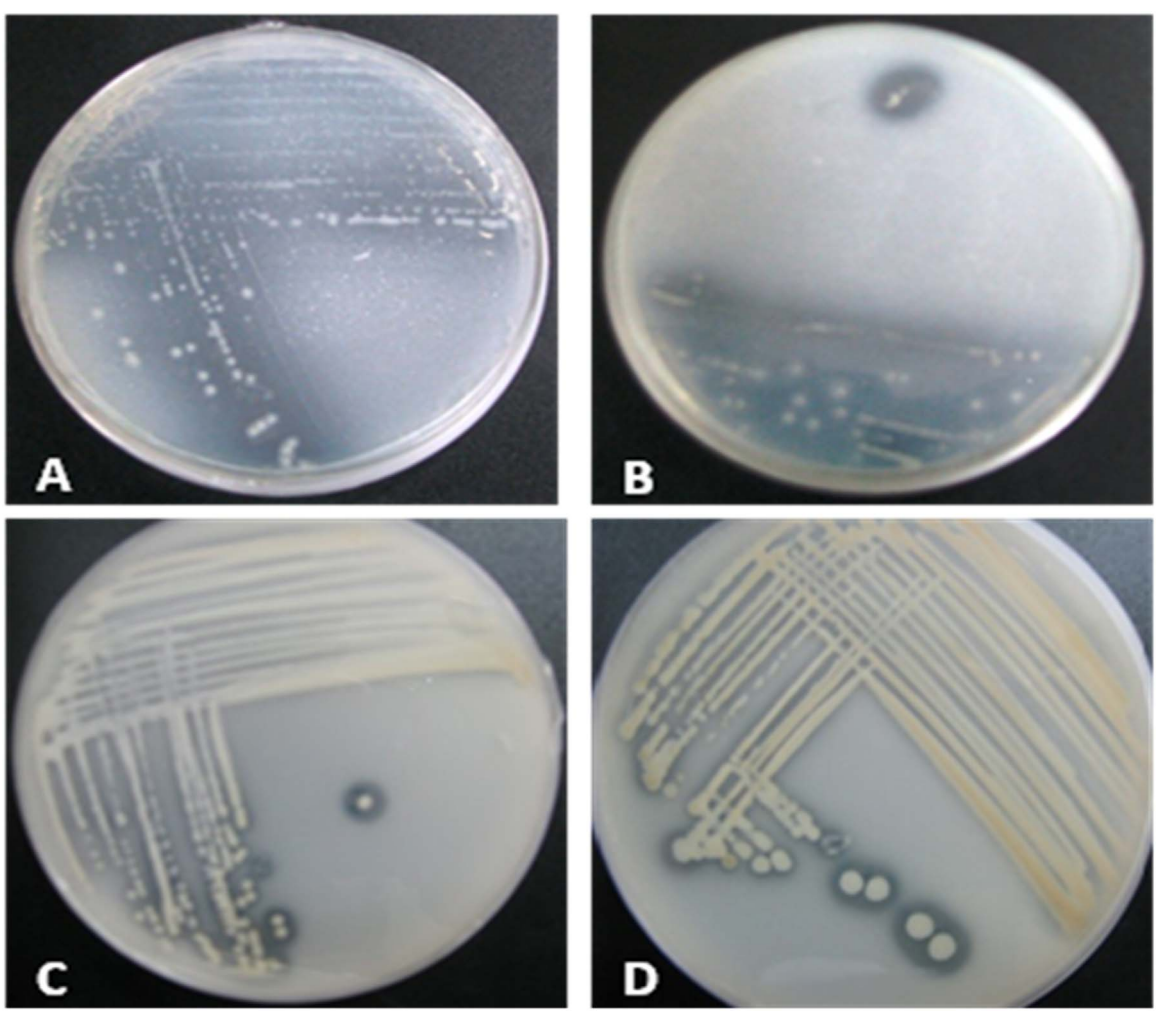

Fig. 3. a, b) NBRIP agar medium plates showing inorganic phosphate solubilization by clear zone formation. C, D) PVK agar medium.

Table 4
Quantification of inorganic phosphorus solubilization by Phosphate solubilizing rhizobacterial isolate.

Samples	Blank	Control	R-1	R-2	R-3	SC-01	SC-03	SC-05	SC-07	SC-08	SC-09	SC-13	SC-15	SC-17	SC-19	SC-20
Control	0.0208															
R-1	−0.0992	−0.225														
R-2	−0.4425	0.4193	0.7902*													
R-3	−0.0074	−0.3466	0.9919*	0.7062*												
SC-01	−0.0191	0.3273	0.847*	0.9951*	0.7729*											
SC-03	−0.1166	−0.3747	0.9877*	0.6846*	0.9995*	0.7534*										
SC-05	−0.0696	−0.4193	0.9789*	0.6484*	0.9969*	0.7206*	0.9988*									
SC-07	0.0195	0.0475	−0.9385*	−0.5301	−0.9747*	−0.6115*	−0.981*	−0.0893								
SC-08	−0.0041	−0.1147	0.9937*	0.8538*	0.9716*	0.9011*	0.964*	0.9499*	−0.8941*							
SC-09	0.0086	0.0123	−0.225	0.4193	−0.3466	0.3273	−0.3747	−0.4193	0.5475	−0.1147						
SC-13	−0.1263	0.2125	0.0987	0.1789	0.8136*	0.0043	0.0191	0.7902*	−0.0021	0.1421	0.2125					
SC-15	−0.1161	0.2235	0.0087	0.0089	0.8006*	0.0193	0.1203	0.7102*	−0.0921	0.2121	0.2235	0.3212				
SC-17	−0.0001	0.2123	0.8712*	0.0712	0.8112*	0.1043	0.1092	0.0702	−0.0011	0.1011	0.4225	0.7261*	0.0032			
SC-19	−0.0163	0.2134	0.8123*	0.1289	0.0021	0.1009	0.1093	0.0902	−0.0623	0.2401	0.2025	0.0623	0.0012	0.8562*		
SC-20	−0.0103	0.2153	0.0198	0.9734*	0.3421	0.1103	0.0191	0.1002	−0.0921	0.0011	0.2315	0.3412	0.1257	0.3204	0.0237	
SC22	−0.3123	0.2541	0.8017*	0.9123*	0.5221	0.1329	0.1092	0.0002	−0.1321	0.0101	0.2115	0.6732*	0.6723*	0.0232	0.9856*	0.6921*

* = Significant at 5% probability level.

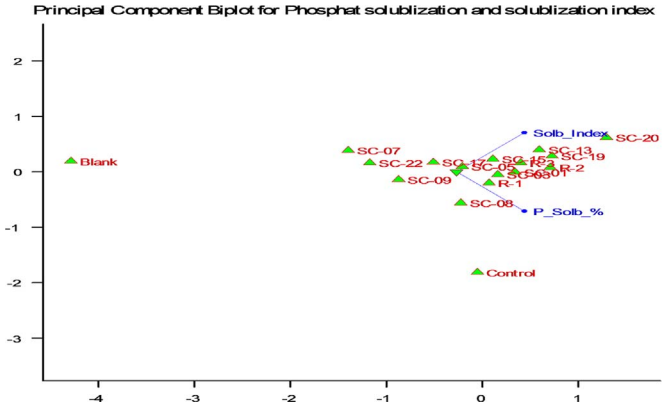

Fig. 5. SI of various PSB isolates.

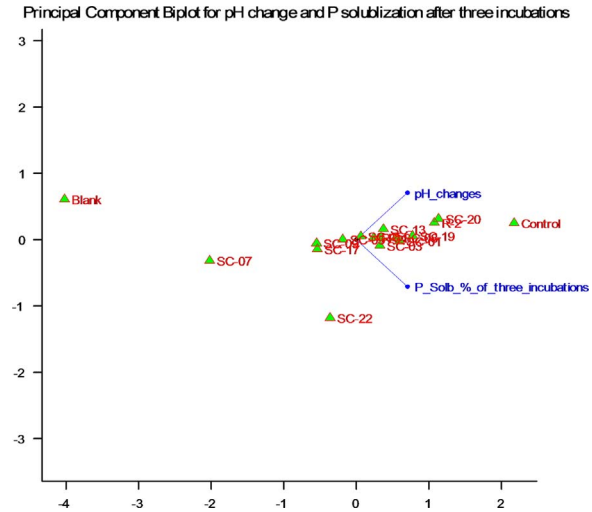

Fig. 6. Final pH of NBRIP medium by various PSB isolates after 3 days incubation.

et al., 2007). The isolates from overnight grown cultures were spread on the agar plates of Pikovskaya's and NBRIP media. The morphology of the colonies was recorded 24 h post inoculation. Further, gram nature of the isolates was studied by the Vincent method. The stained smear was observed for Gram's nature, genera and shape of isolated rhizobacteria under low objective light microscope (Olympus CX31). Further, biochemical tests were performed for identification of phosphate solubilizing rhizobacterial isolates. These tests include Oxidase test, Catalase test and Lactose test (Yasmin and Asghari, 2011) to determine oxidase enzyme, catalase enzyme and lactose enzyme in phosphate solubilizing rhizobacterial isolates. Gram negative bacterial isolates were rapidly identified using Quick Test Strip (QTS) from Miniaturized Identification System-QTS 24 (Desto Laboratories, Karachi) where reactions supplemented with Cytochrome Oxidase identify gram negative bacteria (Panhwar et al., 2014)

2.2. Molecular characterization of isolated strains

Total DNA was extracted by using 24 h old bacterial culture in LB media. Three ml of each culture was centrifuged at 130,00 rpm for 5 min at 4 °C and pellet was washed twice with distilled water. 400 µl CTAB buffer (2% CTAB, 2 µl/ml β-marceptoethanol and 50 µl/ml proteinase K along with 150 µl of lysis buffer (10% SDS and 5 M NaOH)) was added to each tube. Samples were mixed by inverting tubes 2–3 times. Lysis was done by incubation of tubes at 65 °C for 2 h followed by addition of 500 µl of each phenol: chloroform: isoamyl alcohol (25:24:1) and centrifuged for 15 min. Supernatant was then transferred to a fresh 1.5 ml tube and 300 µl of chilled isopropanol was

Table 5
Correlation for intrinsic antibiotic resistance of various Phosphate solubilizing rhizobacterial isolates.

Samples	Blank	Control	R-1	R-2	R-3	SC-01	SC-03	SC-05	SC-07	SC-08	SC-09	SC-13	SC-15	SC-17	SC-19	SC-20
Control	0.0076															
R-1	−0.0058	0.099														
R-2	−0.1019	0.262	0.070													
R-3	−0.0194	0.197	0.199	0.178												
SC-01	−0.0332	0.144	0.197	0.084	0.599*											
SC-03	−0.0471	0.0007	0.094	0.090	0.237	0.999*										
SC-05	−0.238	0.003	0.013	0.029	0.832*	0.085	0.868*									
SC-07	−0.3778	0.315	0.134	0.162	0.378	0.410	0.442	0.828*								
SC-08	−0.1033	0.361	0.169	0.154	0.457	0.012	0.589*	0.930*	0.565*							
SC-09	−0.0194	0.217	0.999*	0.178	−0.011	0.232	−0.120	0.832*	0.378	0.077						
SC-13	−0.0194	0.002	0.009	0.278	0.102	−0.099	−0.132	−0.012	0.378	0.001	0.642*					
SC-15	−0.1033	0.026	0.019	0.054	−0.017	0.014	0.238	−0.030	0.565	0.123	0.400	0.177				
SC-17	−0.3552	0.045	0.475	−0.017	0.505	0.0001	0.563*	0.098	0.000	0.800*	0.505*	0.013	0.075			
SC-19	0.0076	0.067	0.999*	−0.002	−0.007	0.002	0.006	0.093	0.315	−0.022	0.002	0.023	0.061	0.445		
SC-20	−0.1158	0.051	0.012	−0.119	0.110	0.023	0.023	0.141	0.591*	0.192	0.142	0.821*	0.001	0.699*	0.951*	
SC22	−0.1297	0.056	0.013	0.009	0.019	0.680*	−0.009	0.136	0.579*	−0.018	0.012	0.782*	0.102	0.68*	0.956*	0.999*

* = Significant at 5% probability level.

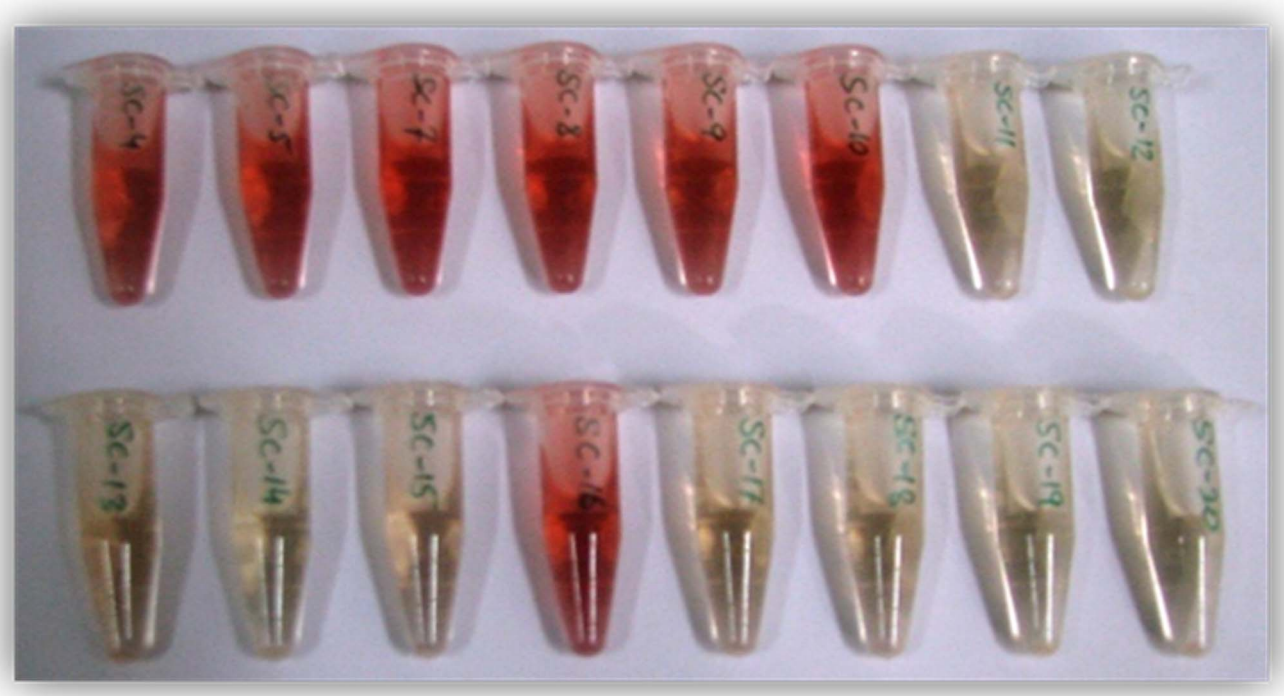

Fig. 7. Red color indicate the presence of Indole 3-acetic Acid. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

added and incubated at −20 °C for 30 min to precipitate DNA. Pellet was washed with 200–400 µl of 70% ethanol. Air dried pellet was re-suspended in 1X TE buffer. The isolated DNA was used to amplify 16 S rRNA gene which was done in a thermal cycler (PCR System 9700; Applied Bio-systems) in 20 µl reaction mixture containing 0.1 mM of each primer, 2.5 mM $MgCl_2$, 1 mM of each dNTPs, 50 ng/µl (2 µl) template DNA and 2.5U Taq DNA polymerase (Thermoscientific). To amplify 1.5 kb gene fragment, universal primers 5′-AACGCGAA-GAACCTTAC-3′ and 5′-CGGTGTGTACAAGACCC-3′ were used with following conditions; initial denaturation at 95 °C for 5 min followed by 35 cycles with denaturation at 95 °C for 1 min, annealing at 57 °C for 1 min, extension at 72 °C for 1.5 min followed by final extension of 10 min at 72 °C. PCR products were analyzed on 1% agarose gel in 1X TAE buffer with ethidium bromide staining and visualized under UV. DNA gel elution kit (Thermoscientific; K0513) was used for elution and purification of the PCR products from gel. Amplified products were sequenced from Macrogen (Korea). Deduced DNA sequences were aligned with sequences present in Genbank database using BLASTN algorithm.

2.3. Phosphate solubilization

PSB isolates were characterized for their ability to solubilize tri-calcium phosphate $Ca_3\ (PO_4)_2$ by the formation of visible dissolution halos on Pikovskaya's and NBRIP agar medium. SI based on colony diameter and halo zone for each phosphate solubilizing bacterial isolate was calculated. The solubilization index (SI) was calculated after five days growth of pin point inoculation on NBRIP at 28 °C as described originally by. The SI was calculated by following equation.

SI = (Total Zone diameter−bacterial colony diameter)

The quantitative method was used for the estimation of inorganic

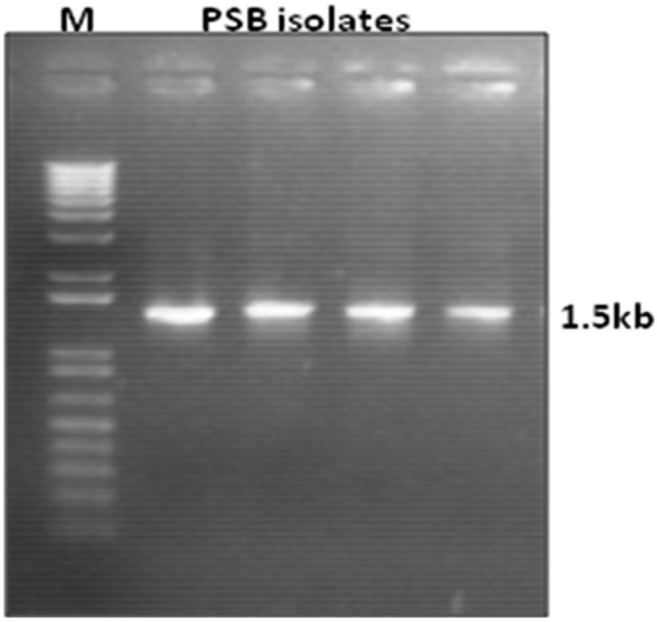

Fig. 8. PCR amplification of 16S rRNA gene using universal primers. Amplification yielded product of 1500 bp (approx). The ladder (L) was 1 Kb (invitrogen life technologies). Lanes 1 and 2 represent the product for PSB isolate R-2, lanes 3 and 4 represent the product for PSB isolate SC-23.

phosphate. For this, pure colonies of all isolates were grown in 50 ml of NBRIP broth with 0.5% Tricholoropropane(TCP) (pH 7.0) and incubated at 28 ± 2 °C for 5 days on a shaking incubator at 180 rpm. Each bacterial strain was inoculated in triplicate. An un-inoculated control was also kept under same conditions. Subsequently, the cultures were centrifuged at 10,000 rpm for 10 min and phosphorus in supernatant was estimated by vanado-molybdate-yellow color method (Mehta and Nautiyal, 2001). To 1 ml aliquot of the supernatant, 5 ml of nitrovanado-molybidic reagent (deionized water, 5% ammonium molybdate solution, 0.25% ammonium vanadate solution, dilute nitric acid) was added and volume was kept at 100 ml. The absorbance was measured at 410 nm in UV/visible spectrophotometer. A 4.3937 g of potassium dihydrogen phosphate in 1000 ml distilled water was used for preparation of the standards of 0.5, 1, 2, 4 and 8 ppm. The total soluble phosphorus was calculated from the regression equation of standard curve. The values of soluble phosphate liberated were expressed in percentages. The pH was also recorded.

2.4. Determination of indole acetic acid (IAA)

A colorimetric method was used for determination of IAA (Illmer and Schinner, 1995). Pure colonies of 15 PSB isolates obtained from NBRIP agar plate were inoculated in 250 ml conical flasks each containing 50 ml LB broth without L-TRP and incubated at 28 ± 2 °C for 3 days at 100 rpm on a shaker. The cultures were then centrifuged at 10,000 rpm for 5 min 1 ml culture supernatant was placed in a test tube and mixed with 2 ml of Salkowski reagent. After 20–25 min, when the color of supernatant containing IAA turned red, the color absorbance was measured at 535 nm using a spectrophotometer. For each test, triplicates samples were taken.

2.5. Intrinsic antibiotic resistance (IAR)

IAR test was carried out to identify the bacterial sensitivity or resistance to four different antibiotics. LB agar plate medium was incubated by spreading 100 µl bacterial culture evenly across the agar surface. Filter paper discs containing different concentration of antibiotics, Amp 25 (Ampicillin 25 µg/ml), K 30 (Kanamycin 30 µg/ml), TE 30 (Tetracycline 30 µg/ml) and S 10 (Streptomycin 10 µg/ml) were placed in each plate. Plates were incubated at 28 ± 2 °C for 3 days. The presence of inhibition zones around the discs of different antibiotics were noted.

2.6. Antifungal activity

The rhizobacterial isolates were screened qualitatively for their ability to suppress growth of the common fungal pathogen, *Fusarium oxysporum*. PDA media was used to study the antagonistic activity of all 15 rhizobacterial isolates. *Fusarium oxysporum* was taken from the peripheral edge of five days old cultures and was placed at the center of the plates. The rhizobacterial isolates were then individually streaked 2.5 cm away from the mycelial plug at 4 opposite locations around the periphery of the plate and the dual culture plates were incubated at 28 ± 2 °C. The zone of inhibition (distance between the edges of the bacterial streak and the fungal mycelium) was recorded after 3 days of growth (Panhwar et al., 2014).

3. Results

The physio-chemical properties of the soil samples depicting electrical conductivity (EC), pH, organic matter available potassium, saturation and available phosphorus in soil samples (Table 1). All rhizobacterial isolates showed good growth on PVK and NBRIP medium. These isolates were identified as PSB based on their ability to solubilize tricalcium phosphate Ca_3 (PO_4) by the formation of visible dilution holes on PVK and NBRIP agar medium. A total of 15 phosphate solubilizing rhizobacterial strains were isolated and purified on the basis of clear zone formation and colony morphology for further analysis (Fig. 1). Rice isolates were designated as R-1, R-2 and R-3 whereas sugarcane isolates were designated as SC-01, SC-03, SC-05, SC-07, SC-08, SC-09, SC-13, SC-15, SC-16, SC-17, SC-19 and SC-22.

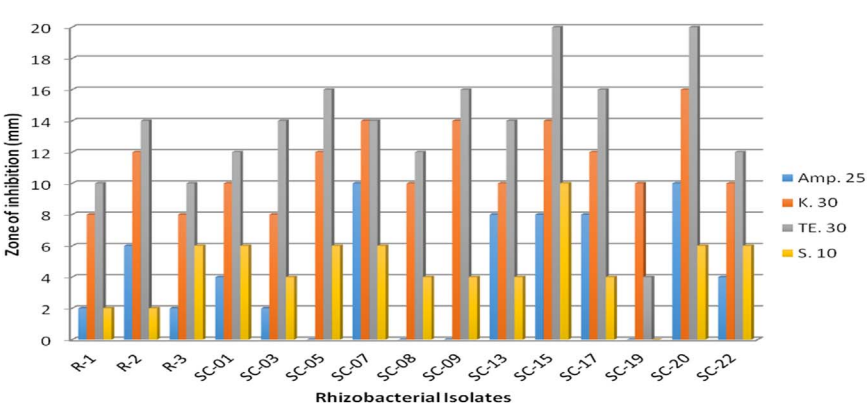

Fig. 9. Intrinsic antibiotic resistance of various Phosphate solubilizing rhizobacterial isolates against different antibiotics: Amp 25 (Ampicillin 25 µg/ml), K 30 (Kanamycin 30 µg/ml), TE 30 (Tetracycline 30 µg/ml) and S 10 (Streptomycin 10 µg/ml).

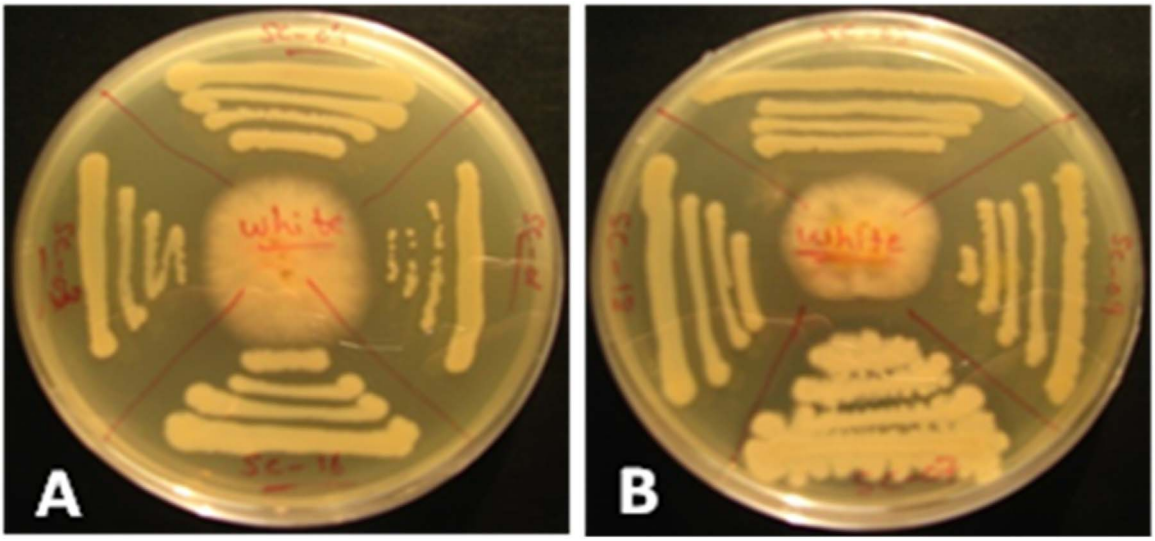

Fig. 10. A-C. PSB isolates showing antagonistic activity against the fungal pathogen, *Fusarium*.

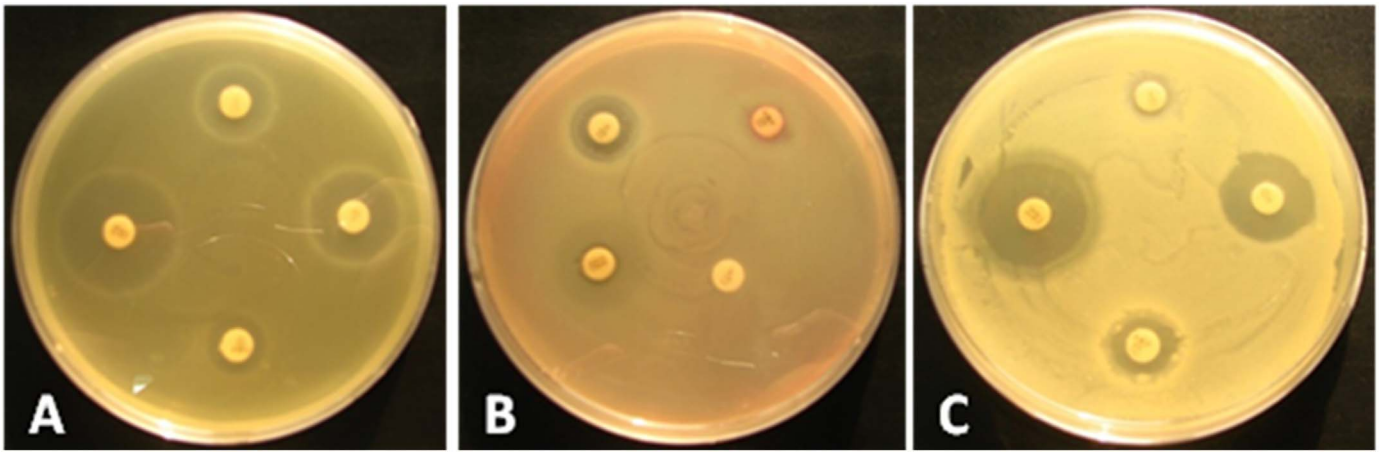

Fig. 11. A-C. Zone of inhibition around antibiotic discs on LB medium.

3.1. Characterization of Rhizobacterial isolates

The 15 PSB genera identified using nomenclature table of QTS-24 were; *Aeromonas hydrophila* (R-1); *Klebsiella pneumoniae* (R-2); *Enterobacter aerogenes* (R-3); *Burkholderia cepacia* (SC-01); *Proteus vulgaris* (SC-03); *Pasteurella multocida* (SC-05); *Stenotrophomonas maltophilia* (SC-07); *Burkholderia mallei* (SC-08); *Burkholderia pseudomallei* (SC-09); *Citrobacter freundii* (SC-13); *Acinetobacter lwoffi* (SC-15); *Pseudomonas fluorescens* (SC-17); *Enterobacter cloacae* (SC-19); *Klebsiella pneumoniae* (SC-20) and *Klebsiella oxytoca* (SC-22).

Morphologically, all 15 rhizobacterial isolates had entire or smooth margins except R-2, SC-01, SC-13, SC-17, SC-19 and SC-22 which have wavy margins with variation in color from skin, yellow to beige. (Table 2). All isolates were gram negative with round in shape (R-2, SC-13, SC-20) while some others were rod shaped (Fig. 1).

Similarly, isolates R-1, SC-01, SC-05, SC-08, SC-09 and SC-17 were found positive for the presence of oxidase enzyme while R-2, R-3, SC-03, SC-07, SC-13, SC-15, SC-19, SC-20 and SC-22 were found negative for oxidase test (Table 3). In parallel, R-1, R-2, R-3, SC-01, SC-05, SC-07, SC-08, SC-09, SC-13, SC-15, SC-17, SC-19, SC-20 and SC-22 were positive for catalase test while only SC-03 was negative for catalase test. In contrast, for lactose test, R-1, R-2, R-3, SC-05, SC-09, SC-13, SC-15, SC-17, SC-19, SC-20 and SC-22 had the ability to ferment lactose while SC-01, SC-03 and SC-07 were negative for this test (Table 3).

Fig. 2 depicts that all 15 rhizobacterial isolates showed variable amount of phosphorus solubilization, Fig. 3 showed clear zone formation around rhizobacterial isolates which is an indication of tricalcium phosphate solubilization. The concentration of inorganic phosphate solubilization by the rhizobacterial isolates ranged from 50.07 to 717.99 ppm (Fig. 4; Table 4). It can be seen that, Fig. 5, showed Solubilization index of all 15 PSB isolates. Fig. 6 showed an inverse relationship between pH value of culture medium and concentration of phosphate solubilized, indicating the organic acid secretion. Difference in the phosphate solubilization was reflected by the change in the pH of the culture medium (Table 5).

Fig. 7 showed that all the rhizobacterial isolates were able to produce variable amount of IAA. Two isolates; R-2 and SC-23 have exhibited best phosphorus solubilization activity and were found best in IAA production. These two were characterized further at molecular level with 1.5 kb 16 SrRNA gene amplification (Fig. 8). The deduced sequences of identified *Klebsiella* sp. (R-2) and *Pseudomonas* sp. (SC-23) were submitted at NCBI database with accession # KF176373 and KF176374.

3.2. Intrinsic antibiotic resistance (IAR)

The intrinsic antibiotic resistance of 15 rhizobacterial isolates against Amp 25, K 30, TE 30 and S 10. It was observed that different rhizobacterial isolates showed different levels of antibiotic resistance. Isolates R-2, SC-07 and SC-20 exhibited higher level of resistance

against all four types of antibiotics. Whereas SC-05, SC-08, SC-09 and SC-19 showed no resistance against tested antibiotics. SC-19 exhibited zero resistance against S 10 alone while the remaining isolates showed intermediate levels of resistance against the antibiotics (Figs. 9 and 10).

3.3. Antagonistic activity

The phosphate solubilizing rhizobacterial isolates were tested qualitatively and it was fond that Out of 15 phosphate solubilizing rhizobacterial, only 7 isolates (SC-01, SC-03, SC-07, SC-09, SC-13, SC-20 and SC-22) showed antifungal activity towards *Fusarium* by induced growth free inhibition zones. After three days of incubation, fungal hyphae of *Fusarium* were unable to reach the bacterial culture and inhibition zone was established with the dimension of inhibition circle (Fig. 11).

Significant positive correlation for phosphate solublization percentage was found among R-1 with R-2, (SC-01, SC-03, SC-08, SC-17, SC-19, SC-22) while negative and significant correlation of R-1 was reported for SC-07. R-2 was significant correlated with R-3, (SC-01, SC-03, SC-05, SC-08, SC-20 and SC-22). R-3 was positively and significantly correlated with R-2, R-1, (SC-01, SC-03, SC-05, SC-08, SC-13, SC-15) and SC17 while negative and significant correlation was found for SC-07. SC-01 and it was significantly and positively correlated with R-2, R-1, R-3, SC-03, SC-05 and SC-08 while negative and significant correlation was reported for SC-07. Strong and significant correlation was recorded for SC-20, SC-19, SC-17 and SC-22. Further significant positive correlation for intrinsic antibiotic resistance of various phosphate solubilizing rhizobacterial isolates was found among R-1 with SC-09 and SC-19. R-3 was positively and significantly correlated with SC-01 and SC-05. SC-01 was significantly and positively correlated with SC-03 and SC-22. SC-05 showed significant correlation with SC-07, SC-08 and SC-09 while SC-07 showed significant correlation with SC-03, SC-08, SC-20 and SC-22. SC-08 showed significant and positive correlation SC-03, SC-05, SC-07 and SC-17 while SC-09 showed significant positive correlation with SC-13 and SC-17. SC-13 was significantly correlated with SC-09, SC-20 and SC22 while SC-17 showed significant correlation for SC-09, SC-20 and SC-22. it was found that strong and significant correlation was recorded for SC-20, SC-19, SC-17 and SC-22.

4. Discussion

Phosphorus is an essential macronutrient for plant growth. A great portion of phosphorus from chemical fertilizers becomes insoluble by its conversion into calcium or magnesium salts in soils and thereby become unavailable to plants. Certain soil microorganisms transform the insoluble forms of phosphorus into soluble forms and thus influence the subsequent availability of phosphate to plant roots (Illmer and Schinner, 1995).

PSB are important components of soil and directly or indirectly influence the soil's health through their useful activities. It is known that rhizospheric microbes mediate many soil processes such as decomposition, nutrient mineralization and nitrogen fixation (Pradhan and Sukla, 2009). In this study, fifteen PSB isolates were selected for Phosphate solubilization activity. All isolates were able to solubilize Phosphorus on NBRIP and pikovskaya media. Perez (Nautiyal, 1999a, 1999b) used NBRIP media for P solubilizing activity and reported similar results. Our results are in agreement with those of Kumar (Perez et al., 2007) since present PSB isolates solubilized tricalcium phosphate.

This study presents a combination of morphological, biochemical and molecular approaches for the identification and characterization of PSB isolates. Morphological approach involved colony and cell morphology, and gram staining; biochemical approach involved oxidase, catalase, and QTS-24; molecular approach involved 16S rRNA gene sequencing for the identification of PSB. All isolates were able to solubilize phosphate after 3 days of incubation. This may be attributed to better growth of PBS and pH drop of the media as suggested by Gen-Fu and Xue-Ping (2005) who observed a linear relationship between soluble phosphorus level in the supernatant and growth of culture. PSB isolates had potential to synthesize IAA. Glick and Pasternak (1998) reported PSB strains isolated from aerobic rice that had prominent effects on plant growth and development accompanied by the production of IAA. Woo et al. (2010) isolated PSB strains from the rhizosphere of Chinese cabbage which were found to be capable of solubilizing phosphorous in the media and besides this, they were also able to produce IAA. Dual culture technique on PDA plates indicated that present PSB strains also had an antagonistic effect on the fungus Fusarium. Kumar et al. (2010) have previously reported similar results that PSB strains (*Pseudomonas* spp.) exhibited the best antagonistic properties against the pathogenic fungus, *Rhizoctonia solani* among all the isolates collected from paddy locations in Seberang Perai, Penang, Malaysia (Woo et al., 2010; Noor et al., 2011).

It was suggested from present study that there was strong and significant correlation among R-1, R-2, R-3, SC-01, SC-03, SC-05, SC-7, SC-20, SC-19, SC-17 and SC-22 for phosphate solublization percentage which indicated that the combination of different isolates may be helpful to improve phosphate solublization in plant to improve yield and potential of crop plants. Selection of PSB isolates may be helpful to efficiency of plant to use and absorb large amount of phosphates. It was found that there was strong correlation among the phosphate solubilizing rhizobacterial isolates including R-1, R-3, SC-01, SC-03, SC-07, SC-09, SC-13, SC-19, SC-17 SC-20 and SC-22 showed antifungal activity towards *Fusarium* by induced growth free inhibition zones. The strong correlation suggested that the combination isolates may be more effective for antifungal activity as compared with single or individual application of phosphate solubilizing rhizobacterial isolates (Xiao et al., 2011; Colwel, 1963; Awaan et al., 2014).

5. Conclusion

In conclusion, 15 PSB were isolated from 8 different locations at Lahore, Sheikhupura, and Gujranwala. The PSB isolates varied in their Phosphate solubilizing efficiency, IAA production, intrinsic antibiotic resistance, and antagonistic activity against the fungal pathogen, Fusarium. It was concluded from correlation studies that combine application of rhizobacterial isolates may be used with good antifungal activity towards *Fusarium* and higher ability to solubilize insoluble phosphorus. Complete spectrum of phytohormones and secondary metabolites produced by these PSB require further study. Present results can provide basis for the large scale isolation and utilization as biofertilizer of more efficient PSB for commercial production of inoculum.

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